Appendix 1

A. 2019 Novel Coronavirus Disease (COVID-19) Diagnostic criteria 5th edition (5 Feb 2020)

Other provinces except Hubei province

1. Suspected cases by clinical criteria

Comprehensive screening of the following epidemiological information and clinical manifestations:

1) Epidemiological information

a) Cases reporting a history of travel to or reside in Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

b) Cases have been exposed to patients with COVID-19 (confirmed by laboratory diagnosis of SARS-CoV-2 RNA detection) within 14 days prior to onset of illness.

c) Cases have been exposed to patients with fever or respiratory symptoms from Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

d) Has contact history with a cluster.

2) Clinical manifestations

a) Fever and / or respiratory symptoms

b) Computed tomography (CT) changes for viral pneumonia

c) Normal or below normal absolute leukocyte count, or low absolute lymphocyte count below normal range in the early stages of the disease. Diagnosis of COVID-19 is made if 1) cases report any one of above epidemiological information and harbor any two of the clinical manifestations, or 2) cases without the above epidemiological information, but harbor all of the above clinical manifestations.

2. Laboratory confirmed cases

Suspected cases with any one of the following:

1) Respiratory tract or blood specimen test positive for SARS-CoV-2 by real-time reverse transcriptase polymerase chain reaction (RT-PCR) assay.

2) Respiratory tract or blood specimen showing high homology with SARS-CoV-2 by genetic sequencing.

Hubei province

1. Suspected cases

Comprehensive screening of the following epidemiological information and clinical manifestations:

1) Epidemiological information

a) Cases reporting a history of travel to or reside in Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

b) Cases have been exposed to patients with COVID-19 (confirmed by laboratory diagnosis of SARS-CoV-2 RNA detection) within 14 days prior to onset of illness.

c) Cases have been exposed to patients with fever or respiratory symptoms from Wuhan city and surrounding areas, or other communities with reported cases within 14 days prior to onset of illness.

d) Has contact history with a cluster.

2) Clinical manifestations

a) Fever and / or respiratory symptoms

b) Normal or below normal absolute leukocyte count, or low absolute lymphocyte count below normal range in the early stages of the disease.

Diagnosis of COVID-19 is made if 1) cases report any one of above epidemiological information and harbor any two of the clinical manifestations, or 2) cases without the above epidemiological information, but harbor any two of clinical manifestations

2. Clinically diagnosed cases

CT chest changes that are consistent with atypical viral pneumonia.

3. Laboratory confirmed cases

Suspected cases or clinically diagnosed cases with any one of the following:

1) Respiratory tract or blood specimen test positive for SARS-CoV-2 by RT-PCR assay.
2) Respiratory tract or blood specimen showing high homology with SARS-CoV-2 by genetic sequencing.

B. National Health Commission Guideline on Severity Classification of COVID-19 (China)

Type	Criteria
Mild	the clinical symptoms were light, and there was no sign of pneumonia on imaging
Moderate	with fever, respiratory tract and other symptoms, imaging suggests pneumonia
Severe	coincide with any of the following: (1) respiratory distress, respiration rate (RR) ≥ 30 times / min; (2) the oxygen saturation ≤ 93% in the resting state; (3) PaO_2 / FiO_2 ≤ 300 mmHg (1mmHg = 0.133 kPa)
Critically ill	coincide with any of the following: (1) respiratory failure occurs and mechanical ventilation is required; (2) shock; (3) the patient develops other organ failure and needs ICU monitoring and treatment

(A and B released in The Diagnosis and Treatment Protocol for Novel Coronavirus Pneumonia, Version 5, National Health Commission, China)

C. IDSA/ATS Guidelines on Severe Pneumonia 2007

2007 Infectious Diseases Society of America/American Thoracic Society Criteria for Defining Severe Community-acquired Pneumonia
Validated definition includes either one major criterion or three or more minor criteria
Minor criteria
Respiratory rate ≥ 30 breaths/min
Pa_{O2}/Fi_{O2} ratio ≤ 250
Multilobar infiltrates
Confusion/disorientation
Uremia (blood urea nitrogen level ≥ 20 mg/dl)
Leukopenia* (white blood cell count < 4,000 cells/μl)
Thrombocytopenia (platelet count < 100,000/μl)
Hypothermia (core temperature < 36°C)
Hypotension requiring aggressive fluid resuscitation
Major criteria
Septic shock with need for vasopressors
Respiratory failure requiring mechanical ventilation

*Due to infection alone (i.e., not chemotherapy induced).

Metlay et al. Diagnosis and Treatment of Adults with Community-acquired Pneumonia. An Official Clnical Practice Guideline of the American Thoracic Society and Infectious Diseases Society of America. AJRCCM 2019.

D. Search criteria

MeSH terms for Pubmed/Medline (also used for google scholar search):
COVID-19, coronavirus, nCoV, SARS-CoV-2, characteristics, comorbidity, mortality, death, risk, oncology, cancer, malignancy, neoplasm, tumor, tumour, and carcinoma.

Search strategy on Pubmed:
(COVID-19 OR coronavirus OR nCoV OR SARS-CoV-2) AND (characteristics OR comorbidity OR mortality OR death OR risk) AND (oncology OR cancer OR malignancy OR neoplasm OR tumor OR tumour OR carcinoma)
Filtered for English language, full article and date range November 1, 2019 to December 31, 2020

E. PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1-2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N.A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4-5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4, appendix 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4-5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4-5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4-8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Appendix 2
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	4-5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	4-5

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	4-5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	4-5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6, Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6-10, Appendix 2-3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	2, 6-12 Appendix 2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6-12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	6-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6-12
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	6-12, Appendix 3
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	16

Appendix 2

Supplementary Table 1: Case series included in meta-analysis

No.	Author	Title	No.	Type of study	Outcome measured	Quality of study	Hospital	Location
1	Argenziano et al	Characterization and clinical course of 1000 patients with coronavirus disease 2019 in New York: retrospective case series	1000	Single centre retrospective cohort study	Severity, death	7	New York-Presbyterian/Columbia University Irving Medical Center	New York, USA
2	Assaad et al	High mortality rate in cancer patients with symptoms of COVID-19 with or without detectable SARS-COV-2 on RT-PCR	302	Single centre retrospective cohort study	Death	7	Comprehensive Cancer Center of Lyon	Lyon France
3	Benelli et al. 2020	SARS-CoV-2 comorbidity and outcome in hospitalized patients in Crema, Italy.	539	Single centre prospective cohort study	Severity	7	General Hospital of Crema	Crema, Italy
4	Cai et al. 2020	COVID-19 in a designated infectious diseases hospital outside Hubei Province, China.	298	Single centre retrospective cohort study	Severity	7	Third People's Hospital of Shenzhen	Shenzhen, China
5	Cao M et al. 2020	Clinical Features of Patients Infected with the 2019 Novel Coronavirus (COVID-19) in Shanghai, China.	198	Single centre retrospective cohort study (not-peer reviewed, pre-print)	Severity	7	Shanghai Public Health Clinical Centre, 20 Jan to 15 Feb 2020	Shanghai, China
6	Cao J et al. 2020	Clinical features and short-term outcomes of 102 patients with coronavirus disease 2019 in Wuhan, China.	102	Single centre retrospective cohort study	Death	7	Wuhan University Zhongnan Hospital in Wuhan, China, 3 Jan to 1 Feb 2020	Wuhan, Hubei, China
7	Chen J et al. 2020	Clinical progression of patients with COVID-19 in Shanghai, China.	249	Single centre retrospective cohort study	Severity	7	Shanghai Public Health Clinical Centre, 20 Jan to 6 Feb, 2020	Shanghai, China
8	Chen N et al. 2020	Epidemiological and clinical characteristics of 99 cases of 2019 novel coronavirus pneumonia in Wuhan, China: a descriptive study.	99	Single centre retrospective cohort study	Severity	7	Jinyintan Hospital in Wuhan, 1 to 20 Jan 2020	Wuhan, Hubei, China
9	Colaneri et al. 2020	Clinical characteristics of coronavirus disease (COVID-19) early findings from a teaching hospital in Pavia, North Italy, 21 to 28 February 2020.	44	Single centre retrospective cohort study	Severity	6	Fondazione IRCCS Policlinico San Matteo, 21 to 28 Feb 2020	Pavia, Italy
10	Dai et al. 2020	Patients with cancer appear more vulnerable to SARS-COV-2: a multi-center study during the COVID-19 outbreak.	641	Multicentre retrospective matched cohort study	Severity	8	14 hospitals in Wuhan, China, 1 Jan to 24 Feb 2020 (Zhongnan, Tongji, Renmin Hospital etc)	Wuhan, Hubei, China
11	Docherty et al 2020	Features of 20 133 UK patients in hospital with covid-19 using the ISARIC WHO Clinical Characterisation Protocol: prospective observational cohort study	20133	Multi centre prospective cohort study	Severity, death	7	Hospitals in England and Wales, UK	UK
12	Du et al. 2020	Predictors of Mortality for Patients with COVID-19 Pneumonia Caused by SARS-CoV-2: A Prospective Cohort Study.	179	Single centre prospective cohort study	Death	7	Wuhan Pulmonary Hospital, 25 Dec 2019 to 7 Feb 2020	Wuhan, Hubei, China
13	Duanmu et al. 2020	Characteristics of Emergency Department Patients With COVID-19 at a Single Site in Northern California: Clinical Observations and Public Health Implications.	100	Single centre observational cross-section study	Admission from ED, Death	6	Stanford University School of Medicine, 4 to 23 March 2020	California, USA
14	Feng et al. 2020	COVID-19 with Different Severity: A Multi-center Study of Clinical Features	476	Single centre retrospective cohort study	Severity and death	7	3 hospitals in Wuhan, Shanghai and Anhui, 1 Jan to 15 Feb 2020	China
15	Guan et al. 2020	Clinical Characteristics of Coronavirus Disease 2019 in China.	1099	Multicentre retrospective cohort study	Severity and death	7	552 hospitals across China, December 11, 2019, and January 29 2020	China
16	Guo et al. 2020	Cardiovascular Implications of Fatal Outcomes of Patients With Coronavirus Disease 2019 (COVID-19)	187	Single centre retrospective cohort study	Severity and death	7	Seventh Hospital of Wuhan City, from 23 Jan to 23 Feb 2020	Wuhan, Hubei, China
17	Huang C et al. 2020	Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China.	41	Single centre prospective cohort study	Severity	7	Jinyintan Hospital till 2 Jan 2020	Wuhan, Hubei, China
18	Huang Y et al. 2020	Clinical characteristics of laboratory confirmed positive cases of SARS-CoV-2 infection in Wuhan, China: A retrospective single center analysis.	34	Single centre retrospective cohort study	Severity	6	Zhongnan Hospital, 21 Dec to 28 Jan 2020	Wuhan, Hubei, China
19	Kim et al. 2020	Clinical Course and Outcomes of Patients with Severe Acute Respiratory Syndrome Coronavirus 2 Infection: a Preliminary Report of the First 28 Patients from the Korean Cohort Study on COVID-19.	28	Multicentre retrospective cohort study	Severity	6	11 Hospitals in Korea	Korea
20	Kuderer et al. 2020	Clinical impact of COVID-19 on patients with cancer (CCC19): a cohort study	928	Multicentre retrospective cohort study	Severity, death	7	US, Canada, Spain Hospitals	US, Canada, Spain
21	Lee LY et al.	COVID-19 prevalence and mortality in patients with cancer and the effect of primary tumour subtype and patient demographics: a prospective cohort study	800	Multicentre prospective cohort study	Severity, death	7	UK Hospitals	UK
22	Li et al. 2020	Risk factors for severity and mortality in adult COVID-19 inpatients in Wuhan.	548	Single centre retrospective cohort study	Severity and death	7	Tongji Hospital, 26 Jan to 5 Feb 2020	Wuhan, Hubei, China
23	Lian et al. 2020	Analysis of Epidemiological and Clinical features in older patients with Corona Virus Disease 2019 (COVID-19) out of Wuhan.	788	Multicentre retrospective cohort study	Severity	8	Zhejiang Province, 17 Jan to 12 Feb 2020	Wuhan. Hubei, China
24	Liang et al. 2020	Cancer patients in SARS-CoV-2 infection: a nationwide analysis in China.	1590	Multicentre retrospective cohort study	Severity and death	8	575 hospitals in China as of 31 Jan 2020	China
25	Liu et al. 2020	Clinical characteristics of novel coronavirus cases in tertiary hospitals in Hubei Province.	137	Multicentre retrospective cohort study	Death	7	9 hospitals in Hubei, 30 Dec 2019 to 24 Jan 2020 (Tongji Hosp, other hospitals outside Wuhan)	Hubei, China
26	Ma et al. 2020	Clinical characteristics and prognosis in cancer patients with COVID-19: A single center's retrospective study.	1380	Single centre retrospective cohort study	Severity and death	8	Renmin Hospital of Wuhan University, 1 Jan to 30 March 2020	Wuhan, Hubei, China
27	Mehta et al. 2020	Case Fatality Rate of Cancer Patients with COVID-19 in a New York Hospital System.	218	Single centre retrospective cohort study (peer-reviewed, pre-print)	Death	7	Montefiore Medical Center and Albert Einstein College of Medicine, 18 March to 8 April 2020	New York, USA
28	Miyashita et al. 2020	Do Patients with Cancer Have a Poorer Prognosis of COVID-19? An Experience in New York City.	5688	Single centre retrospective cohort study	Severity and death	8	Mount Sinai Health System, 1 March to 6 April 2020	New York, USA
29	Myers et al. 2020	Characteristics of Hospitalized Adults With COVID-19 in an Integrated Health Care System in California.	377	Multicentre retrospective cohort study	Severity	7	Kaiser Permanente Northern California (KPNC) health system, 1 to 31 March 2020	California, USA
30	Nikpouraghdam et al. 2020	Epidemiological characteristics of coronavirus disease 2019 (COVID-19) patients in IRAN: A single center study.	2964	Single centre retrospective cohort study	Death	7	Baqiyatallah Hospital, 19 Feb to 15 April 2020	Tehran, Iran
31	Pan et al. 2020	Clinical Characteristics of COVID-19 Patients With Digestive Symptoms in Hubei, China: A Descriptive, Cross-Sectional, Multicenter Study	204	Multicentre retrospective cross sectional study	Severity and death	7	3 hospitals in Hubei province, including Wuhan Hanan Hospital, Wuhan	Hubei, China

							Union Hospital, and Huanggang Central Hospital, 18 Jan to 28 Feb 2020	
32	Paranjpe et al. 2020	Clinical Characteristics of Hospitalized Covid-19 Patients in New York City.	1078	Multicentre retrospective cohort study (not peer reviewed, pre-print)	Death	8	Mount Sinai Health System Hospitals, 27 Feb to 2 April 2020	New York, USA
33	Petrilli et al 2020	Factors associated with hospital admission and critical illness among 5279 people with coronavirus disease 2019 in New York City: prospective cohort study	5279	Single centre prospective cohort study	Severity, death	7	NYU Langone Health	New York, USA
34	Pinato et al. 2020	Clinical Portrait of the SARS-CoV-2 Epidemic in European Patients with Cancer	890	Multicentre retrospective cohort study	Severity, death	7	Multiple Hospitals in Europe	Europe
35	Robilotti et al. 2020	Determinants of COVID-19 disease severity in patients with cancer	423	Single centre retrospective cohort study	Severity, death	7	MSKCC	New York, USA
36	Richardson et al. 2020	Presenting Characteristics, Comorbidities, and Outcomes Among 5700 Patients Hospitalized With COVID-19 in the New York City Area.	5700	Multicentre retrospective cohort study	Severity and death	8	Northwell Health Hospitals, 1 March to 4 April 2020	New York, USA
37	Shi H et al. 2020	Radiological findings from 81 patients with COVID-19 pneumonia in Wuhan, China: a descriptive study.	81	Multicentre retrospective cohort study	Not defined	6	Wuhan Jinyintan hospital or Union Hospital of Tongji Medical College, 20 Dec to 23 Jan 2020	Wuhan, Hubei, China
38	Shi Y et al. 2020	Host susceptibility to severe COVID-19 and establishment of a host risk score: findings of 487 cases outside Wuhan.	487	Multicentre retrospective cohort study	Not defined	7	Hospitals in Zhejiang Province, until 17 Feb 2020	Zhejiang, China
39	Tomlins et al. 2020	Clinical features of 95 sequential hospitalised patients with novel coronavirus 2019 disease (COVID-19), the first UK cohort	95	Single centre retrospective cohort study	Severity and death	7	North Bristol NHS Trust, 10 to 30 March 2020	UK
40	Wan et al. 2020	Clinical features and treatment of COVID-19 patients in northeast Chongqing.	135	Single centre retrospective cohort study	Not defined	7	Chongqing University Three Gorges Hospital, 23 Jan to 8 Feb 2020	Chongqing, Sichuan, China
41	Wang D et al. 2020	Clinical Characteristics of 138 Hospitalized Patients With 2019 Novel Coronavirus-Infected Pneumonia in Wuhan, China.	138	Single centre retrospective cohort study	Severity	6	Zhongnan Hospital of Wuhan University, 1 to 28 Jan 2020	Wuhan, Hubei, China
42	Wang R et al. 2020	Epidemiological and clinical features of 125 Hospitalized Patients with COVID-19 in Fuyang, Anhui, China.	125	Single centre retrospective cohort study	Severity	7	NO.2 People's Hospital of Fuyang City 20 Jan to 9 Feb 2020.	Anhui, China
43	Wu C et al. 2020	Risk Factors Associated With Acute Respiratory Distress Syndrome and Death in Patients With Coronavirus Disease 2019 Pneumonia in Wuhan, China.	201	Single centre retrospective study	Severity and death	8	Jinyintan Hospital in Wuhan, China, 25 Dec to 26 Jan 2020	Wuhan, Hubei, China
44	Wu J et al. 2020	Clinical Characteristics of Imported Cases of COVID-19 in Jiangsu Province: A Multicenter Descriptive Study.	80	Multicentre retrospective cohort study	Severity	7	Three Grade IIIA hospitals of Jiangsu, 22 Jan to 14 Feb 2020	Jiangsu, China
45	Xu et al. 2020	Imaging and clinical features of patients with 2019 novel coronavirus SARS-CoV-2.	90	Single centre retrospective study	Severity	7	Guangzhou Eighth People's Hospital, 23 Jan to 4 Feb 2020	Guangzhou, China
46	Yang et al. 2020	Clinical characteristics and imaging manifestations of the 2019 novel coronavirus disease (COVID-19):A multi-center study in Wenzhou city, Zhejiang, China.	149	Multicentre retrospective cohort study	Severity	7	Three tertiary hospitals of Wenzhou, 17 Jan to 10 Feb 2020	Wenzhou, Zhejiang, China
47	Yang K et al 2020	Clinical characteristics, outcomes, and risk factors for mortality in patients with cancer and COVID-19 in Hubei, China: a multicentre, retrospective, cohort stud	205	Multientre retrospective cohort study	Severity, death	7	9 Hospitals in Wuhan, 11 Jan to 18 March (overlap with other case series)	Wuhan, China
48	Yarza et al	SARS-CoV-2 infection in cancer patients undergoing active treatment: analysis of clinical features and predictive factors for severe respiratory failure and death	63	Single centre retrospective cohort study	Death	6	Hospital in Madrid	Madrid, Spain
49	Yu et al. 2020	SARS-CoV-2 Transmission in Patients With Cancer at a Tertiary Care Hospital in Wuhan, China.	1524	Single centre retrospective study	Severity and death	6	Zhongnan Hospital, 30 Dec to 17 Feb 2020	Wuhan, Hubei, China
50	Zhang G et al. 2020	Clinical features and short-term outcomes of 221 patients with COVID-19 in Wuhan. China.	221	Single centre retrospective cohort study	Severity	7	Zhongnan Hospital of Wuhan, 2 Jan to 10 Feb 2020	Wuhan, Hubei, China
51	Zhang J et al. 2020	Risk factors for disease severity, unimprovement, and mortality in COVID-19 patients in Wuhan, China.	663	Single centre retrospective cohort study	Severity	7	Renmin Hospital of Wuhan University, 11 Jan to 2 Feb 2020	Wuhan, Hubei, China
52	Zhang L et al. 2020	Clinical characteristics of COVID-19-infected cancer patients: a retrospective case study in three hospitals within Wuhan, China.	1276	Single centre retrospective cohort study	Severity and death	8	Tongji Sino-French New Town Hospital, Union Red Cross Hospital, and Union West Hospital, 13 Jan to 26 Feb 2020	Wuhan, Hubei, China
53	Zhang N et al. 2020	A multicentre study of coronavirus disease 2019 outcomes of cancer patients in Wuhan, China	1548	Multicentre retrospective observational study (peer reviewed, pre-print)	Severity, death	7	Zhongnan Hospital of Wuhan University, the Fifth hospital of Wuhamn, the seventh hospital of Wuhan and Wuhan Hankou hospital, 5 Jan to 18 Feb 2020	Wuhan, Hubei, China
54	Zhang S et al. 2020	Short Term Outcomes and Risk Factors for Mortality in Patients with COVID-19 in Wuhan, China: A Retrospective Study.	315	Single centre retrospective cohort study	Death	7	Wuhan Union Hospital West Area, 12 Jan to 3 Feb 2020	Wuhan, Hubei, China
55	Zhao et al. 2020	Clinical characteristics of patients with 2019 coronavirus disease in a non-Wuhan area of Hubei Province, China: a retrospective study.	91	Single centre retrospective cohort study	Severity	7	Jingzhou Central Hospital, 16 Jan to 10 Feb 2020	Jingzhou, Hubei, China
56	Zhou et al. 2020	Clinical course and risk factors for mortality of adult inpatients with COVID-19 in Wuhan, China: a retrospective cohort study.	191	Multicentre retrospective cohort study	Death	7	Jinyintan Hospital and Wuhan Pulmonary Hospital, 29 Dec to 31 Jan 2020	Wuhan, Hubei, China
57	Zhu et al. 2020	Initial clinical features of suspected coronavirus disease 2019 in two emergency departments outside of Hubei, China.	32	Single centre retrospective cohort study	Not defined	6	First Affiliated Hospital of USTC, 24 Jan to 20 Feb 2020	Hefei, China

Supplementary Table 2: Meta-analyses and database reports on COVID-19 (not included in analysis)

No.	Authors	Title:
1.	Desai et al.	COVID-19 and Cancer: Lessons From a Pooled Meta-Analysis
2.	Emami et al.	Prevalence of Underlying Diseases in Hospitalized Patients with COVID-19: a Systematic Review and Meta-Analysis
3.	Wang B et al.	Does comorbidity increase the risk of patients with COVID-19: evidence from meta-analysis
4.	Cao Y et al.	Imaging and clinical features of patients with 2019 novel coronavirus SARS-CoV-2: A systematic review and meta-analysis
5.	Hu Y et al.	Prevalence and severity of corona virus disease 2019 (COVID-19): A systematic review and meta-analysis
6.	Yang J et al.	Prevalence of comorbidities and its effects in patients infected with SARS-CoV-2: a systematic review and meta-analysis
7.	Fu L et al.	Clinical characteristics of coronavirus disease 2019 (COVID-19) in China: A systematic review and meta-analysis
8.	Li LQ et al.	COVID-19 patients' clinical characteristics, discharge rate, and fatality rate of meta-analysis
9.	Zhu J et al.	Clinicopathological Characteristics of 8697 Patients With COVID-19 in China: A Meta-Analysis
10.	Zheng Z et al.	Risk factors of critical & mortal COVID-19 cases: A systematic literature review and meta-analysis
11.	Rodriguez-Morales et al.	Clinical, laboratory and imaging features of COVID-19: A systematic review and meta-analysis
12.	Ofori-asenso et al.	Cancer is associated with severe disease in COVID-19 patients: a systematic review and meta-analysis
13.	Khan et al.	Effects of underlying morbidities on the occurrence of deaths in COVID- 19 patients: A systematic review and meta-analysis (in-press)
14.	Moujaess et al.	Cancer patients and research during COVID-19 pandemic: A systematic review of current evidence
15.	Sun P et al.	Clinical characteristics of hospitalized patients with SARS-CoV-2 infection: A single arm meta-analysis
16.	Jin JM et al.	Gender Differences in Patients with COVID-19 Focus on Severity and Mortality
17.	Deng G et al.	Clinical determinants for fatality of 44,672 patients with COVID-19
18.	Wu and Mcgoogan	Characteristics of and Important Lessons From the Coronavirus Disease 2019 (COVID-19) Outbreak in China: Summary of a Report of 72 314 Cases From the Chinese Center for Disease Control and Prevention
19.	Onder G, Rezza G, Brusaferro S.	Case-Fatality Rate and Characteristics of Patients Dying in Relation to COVID-19 in Italy
20.	Guan et al	Comorbidity and its impact on 1590 patients with Covid-19 in China: A Nationwide Analysis
21.	Adams ML, Katz DL, Grandpre J.	Population-Based Estimates of Chronic Conditions Affecting Risk for Complications from Coronavirus Disease, United States
22.	Kang YJ et al	Mortality Rate of Infection With COVID-19 in Korea From the Perspective of Underlying Disease
23.	Barek et al	Impact of age, sex, comorbidities and clinical symptoms on the severity of COVID-19 cases: A meta-analysis with 55 studies and 10014 cases.
24.	Fuentes-Antrás J	A snapshot of COVID-19 infection in patients with solid tumors.
25.	Sng C et al	Cancer History and Systemic Anti-Cancer Therapy Independently Predict COVID-19 Mortality: A UK Tertiary Hospital Experience.
26.	Pinto et al	SARS-CoV-2 Positive Hospitalized Cancer Patients during the Italian Outbreak: The Cohort Study in Reggio Emilia. Biology (Basel).
27.	Sun L et al	Rates of COVID-19-related Outcomes in Cancer compared to non-Cancer Patients. (Pre-print)
28.	Anantharaman et al	SARS-CoV-2 Clinical Outcomes in Patients with Cancer in a Large Integrated Health Care System in Northern California.
29.	Zarifkar P et al	Clinical Characteristics and Outcomes in Patients with COVID-19 and Cancer: a Systematic Review and Meta-analysis.
30.	Jarahzadeh MH et al	Cancer and Coronavirus Disease (COVID-19): Comorbidity, Mechanical Ventilation, and Death Risk.
31.	ElGohary GM et al	The risk and prognosis of COVID-19 infection in cancer patients: A systematic review and meta-analysis.
32.	Russell B et al	Factors Affecting COVID-19 Outcomes in Cancer Patients: A First Report From Guy's Cancer Center in London.
33.	Duarte MBO et al	Outcomes of COVID-19 Patients under Cytotoxic Cancer Chemotherapy in Brazil.
34.	Gagandeep Brar et al	COVID-19 Severity and Outcomes in Patients With Cancer: A Matched Cohort Study
35.	Yang B et al	Risk of Coronavirus Disease Incidence, Severe Presentation, and Mortality in Patients with Lung Cancer.
36.	Motamedi F et al	Clinical characteristics, outcomes, and risk factors for mortality in hospitalized patients with COVID-19 and cancer history: a propensity score-matched study.
37.	Cheruiyot I et al	Cancer is associated with coronavirus disease (COVID-19) severity and mortality: A pooled analysis.

38.	Antikchi MH et al	The Risk and Prevalence of COVID-19 Infection in Colorectal Cancer Patients: a Systematic Review and Meta-analysis.
39.	Hultcrantz M et al	COVID-19 infections and outcomes in patients with multiple myeloma in New York City: a cohort study from five academic centers.
40.	Fillmore NR et al	Prevalence and outcome of COVID-19 infection in cancer patients:a national Veterans Affairs study.
41.	Venkatesulu BP et al	A systematic review and meta-analysis of cancer patients affected by a novel coronavirus.
42.	Joharatnam-Hogan et al	Outcomes of the 2019 novel coronavirus in patients with or without a history of cancer: a multi-centre North London experience.
43.	Alpert N et al	Clinical Course of Cancer Patients With COVID-19: A Retrospective Cohort Study.
44.	Ali J et al	COVID-19-positive cancer patients undergoing active anticancer treatment: An analysis of clinical features and outcomes.
45.	Jee J et al	Chemotherapy and COVID-19 Outcomes in Patients With Cancer
46.	Liu H et al	The effect of anticancer treatment on cancer patients with COVID-19: A systematic review and meta-analysis.
47.	Park R et al	Sex-bias in COVID-19-associated illness severity and mortality in cancer patients: A systematic review and meta-analysis.
48.	Asokan I et al	The COVID-19 Pandemic and its Impact on the Cardio-Oncology Population.
49.	Rogado J et al	Characteristics and outcome of lung cancer patients and Covid-19 infection in a second-level hospital during the first 3 months of the pandemic.
50.	Park R et al	Association of active oncologic treatment and risk of death in cancer patients with COVID-19: a systematic review and meta-analysis of patient data

The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses

COHORT STUDIES

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Outcome categories. A maximum of two stars can be given for Comparability

Selection

1) Representativeness of the exposed cohort
   a) truly representative of the average cancer COVID-19 cases in the community *
   b) somewhat representative of the average cancer COVID-19 cases in the community *
   c) selected group of users eg nurses, doctors
   d) no description of the derivation of the cohort

2) Selection of the non-exposed cohort
   a) drawn from the same community as the exposed cohort *
   b) drawn from a different source
   c) no description of the derivation of the non-exposed cohort

3) Ascertainment of exposure
   a) secure record (eg medical records) *
   b) structured interview *
   c) written self-report
   d) no description

4) Demonstration that outcome of interest was not present at start of study
   a) yes *

b) no

Comparability

5) Comparability of cohorts on the basis of the design or analysis
a) study controls for median age *
b) study controls for any additional factors such as other co-morbidities *

Outcome

6) Assessment of outcome
a) independent blind assessment *
b) record linkage *
c) self-report
d) no description

7) Was follow-up long enough for outcomes to occur
a) yes (until discharge or death or at least 1 month, whichever come first) *
b) no

8) Adequacy of follow up of cohorts
a) complete follow up - all subjects accounted for *
b) subjects lost to follow up unlikely to introduce bias - small number lost - > 80 % follow up, or description provided of those lost) *
c) follow up rate < 60 % and no description of those lost
d) no statement

Case-control studies

1) Is the case definition adequate?
a) yes, with independent validation *
b) yes, eg record linkage or based on self reports
c) no description

2) Representativeness of the cases
a) consecutive or obviously representative series of cases *
b) potential for selection biases or not stated

3) Selection of Controls
a) community controls *
b) hospital controls
c) no description

4) Definition of Controls
a) no history of disease (endpoint) *
b) no description of source

Comparability

5) Comparability of cases and controls on the basis of the design or analysis
a) study controls for median age *
b) study controls for any additional factor * (co-morbidities)

Exposure

6) Ascertainment of exposure
a) secure record (eg surgical records) *
b) structured interview where blind to case/control status *
c) interview not blinded to case/control status
d) written self report or medical record only
e) no description

7) Same method of ascertainment for cases and controls
a) yes *

b) no

8) Non-Response rate
a) same rate for both groups *
b) non respondents described
c) rate different and no designation

Supplementary Table 3: The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses

N	Authors	Selection				Comparability		Outcome			Study quality
	Cohort studies	1 Exposed cohort truly representative	2 Non-exposed cohort drawn from same community	3 Ascertainment of exposure	4 Outcome of interest not present at start	5A Cohorts comparable ie. Median age	5B Cohorts comparable on other co-morbidities	6 Quality of outcome assessment	7 Follow-up long enough for outcomes to occur	8 Complete accounting for cohorts	
	Case control studies	1 Case definition adequate	2 Representativeness with the cases	3 Selection of controls	4 Definition of controls	5A Case and control comparable on the basis of analysis ie median age	5B Case and control comparable on other co-morbidities	6 Ascertainment of exposure	7 Same method of ascertainment for cases and controls	8 Non-Response rate	
1	Argenziano et al.	*	*	*	*			*	*	*	7
2	Assaad et al.	*	*	*	*			*	*	*	7
3	Benelli et al.	*	*	*	*			*	*	*	7
4	Cai et al.	*	*	*	*			*	*	*	7
5	Cao M et al.	*	*	*	*			*	*	*	7
6	Cao J et al.	*	*	*	*			*	*	*	7
7	Chen J et al	*	*	*	*			*	*	*	7
8	Chen N et al.	*	*	*	*			*	*	*	7
9	Colaneri et al.	*	*	*	*			*		*	6
10	Dai et al.	*	*	*	*	*		*	*	*	8
11	Docherty et al.	*	*	*	*			*	*	*	7
12	Du et al.	*	*	*	*			*	*	*	7
13	Duanmu et al.	*	*	*	*			*		*	6
14	Feng et al.	*	*	*	*			*	*	*	7
15	Guan et al.	*	*	*	*			*	*	*	7
16	Guo et al.	*	*	*	*			*	*	*	7
17	Huang C et al.	*	*	*	*			*	*	*	7
18	Huang Y et al.	*	*	*	*			*		*	6
19	Kim et al.	*	*	*	*			*		*	6
20	Kuderer et al.	*	*	*	*			*	*	*	7
21	Lee LY et al.	*	*	*	*			*	*	*	7
22	Li et al.	*	*	*	*			*	*	*	7
23	Lian et al.	*	*	*	*	*		*	*	*	8
24	Liang et al.	*	*	*	*		*	*	*	*	8
25	Liu et al.	*	*	*	*			*	*	*	7
26	Ma et al.	*	*	*	*	*		*	*	*	8
27	Mehta et al.	*	*	*	*			*	*	*	7
28	Miyashita et al.	*	*	*	*	*		*	*	*	8
29	Myers et al.	*	*	*	*			*	*	*	7
30	Nikpouraghdam et al.	*	*	*	*			*	*	*	7
31	Pan et al.	*	*	*	*			*	*	*	7
32	Paranjpe et al.	*	*	*	*		*	*	*	*	8
33	Petrilli et al.	*	*	*	*			*	*	*	7
34	Pinato et al.	*	*	*	*			*	*	*	7
35	Robilotti et al.	*	*	*	*			*	*	*	7
36	Richardson et al.	*	*	*	*	*		*	*	*	8
37	Shi H et al.	*	*	*	*			*		*	6
38	Shi Y et al.	*	*	*	*	*		*	*		7
39	Tomlins et al	*	*	*	*			*	*	*	7
40	Wan et al.	*	*	*	*			*	*	*	7
41	Wang D et al.	*	*	*	*			*	*		6
42	Wang R et al.	*	*	*	*			*	*	*	7
43	Wu C et al.	*	*	*	*	*		*	*	*	8
44	Wu J et al.	*	*	*	*			*	*	*	7
45	Xu et al.	*	*	*	*			*	*	*	7
46	Yang et al.	*	*	*	*			*	*	*	7
47	Yang K et al.	*	*	*	*			*	*	*	7
48	Yarza et al.	*	*	*	*			*	*		6
49	Yu et al.	*		*	*			*	*	*	6
50	Zhang G et al.	*	*	*	*			*	*	*	7
51	Zhang J et al.	*	*	*	*			*	*	*	7
52	Zhang L et al.	*	*	*	*		*	*	*	*	8
53	Zhang N et al.	*	*	*	*			*	*	*	7
54	Zhang S et al.	*	*	*	*			*	*	*	7
55	Zhao et al.	*	*	*	*			*	*	*	7
56	Zhou et al.	*	*	*	*			*	*	*	7
57	Zhu et al.	*	*	*	*				*	*	6

Appendix 3

Supplementary Table 1: List of 230 patients with available individual patient data and oncological history

Liang et al.									
No	**Gender**	**Age**	**Co-morbidities**	**Length of dx of cancer in years**	**Malignancy**	**Treatment**	**Severity (Y/N)**	**Alive (Y/N)**	**Stage IV (Y/N)**
1	M	83	No	4	Bladder ca	Postoperative routine follow-up	Y	NA	N
2	M	87	DM, HTN	2	Colon ca	Postoperative routine follow-up	Y	NA	N
3	M	67	No	0	Adrenal neoplasm	3 weeks after operation	Y	NA	N
4	F	68	No	8	Breast ca	Surgical resection plus adjuvant chemotherapy 8 years ago	Y	NA	N
5	M	58	No	7	CRCC	Recurrence, on immunotherapy	Y	NA	Y
6	F	62	No	4	Breast ca	Surgical resection plus adjuvant chemotherapy 4 years ago	Y	NA	N
7	M	56	No	5	Rectal ca	Surgical resection plus adjuvant chemotherapy 5 years ago	N	NA	N
8	M	53	No	16	Colon ca	Surgical resection plus adjuvant chemotherapy 16 years ago	N	NA	N
9	F	63	No	1	Papillary thyroid ca	Postoperative, in TSH inhibition therapy	N	NA	N
10	M	47	No	1	Lung adenoca	In chemotherapy for advanced tumor	N	NA	Y
11	F	52	No	0.5	Breast ca	Postoperative, loss of chemotherapy information	N	NA	N
12	M	47	No	1	Lymphoma	N.A.	N	NA	NA
13	M	80	DM, HTN, CVS	4	Bladder ca	Postoperative, no chemotherapy information	Y	NA	N
14	M	79	COPD	5	Colorectal ca	Surgical resection plus adjuvant radiotherapy 5 years ago	Y	NA	N
15	M	63	No	1	Lung adenoca	In chemotherapy for advanced tumour	Y	NA	Y
16	F	58	CKD	2	Lung carcinoma in situ	Postoperative routine follow-up	N	NA	N
17	M	58	No	2	Lung adenoca	Postoperative, on targeted therapy	N	NA	N
18	F	55	No	6	Lung adenoca	Advanced, on targeted therapy	N	NA	Y
Yu et al.									
No	**Gender**	**Age**	**Co-morbidities**	**Length of dx of cancer**	**Malignancy**	**Treatment**	**Severity (Y/N)**	**Alive (Y/N)**	**Stage IV (Y/N)**
1	M	NA	NA	NA	NSCLC	Concurrent TKI with RT	N	Y	N
2	M	NA	NA	NA	NSCLC	Week 5 of chemoradiotherapy	Y	N	N
3	F	NA	NA	NA	NSCLC	Pleural effusion, BSC	N	Y	Y
4	M	NA	NA	NA	NSCLC	Follow up at 2 year post chemoRT	Y	N	N
5	M	NA	NA	NA	NSCLC	After first cycle of doce, cis and sintilimab	N	Y	Y
6	M	NA	NA	NA	NSCLC	After third cycle of pem/carbo and pembro	N	Y	Y
7	M	NA	NA	NA	NSCLC	Surgery (April 2019) followed by adjuvant doce/cis	N	Y	N
8	M	NA	NA	NA	Rectal ca	BSC	N	Y	Y
9	M	NA	NA	NA	Colon ca	Newly diagnosed	N	Y	NA
10	M	NA	NA	NA	Pancreatic ca	BSC	Y	N	Y
11	F	NA	NA	NA	Breast ca	Adjuvant radiotherapy	N	Y	N
12	M	NA	NA	NA	Urothelial ca	BSC	N	Y	Y
Zhang L et al.									
No	**Gender**	**Age**	**Co-morbidities**	**Length of dx of cancer in months**	**Malignancy**	**Treatment**	**Severity (Y/N)**	**Alive (Y/N)**	**Stage IV (Y/N)**
1	M	56	CHB, cirrhosis	26	Liver ca	Stage IV, recurrence with bone mets, bone RT with 14 days	Y	NA	Y
2	M	74	DM	3	Esophageal ca	Stage IV, chemotherapy 1 month ago	Y	NA	Y
3	M	66	HTN, CHD	11	Lung SCC	Surgical resection plus adjuvant chemoRT 10m ago, immune therapy 3m ago, gem and nivo within 14 days	Y	NA	Y
4	M	65	No	23	Lung adenoca	Stage IV, chemo for advanced tumour. Last cycle of pemetrexed was within 14 days	Y	NA	Y
5	M	65	No	4	Esophageal ca	Stage III, surgical resection plus adjuvant chemo and RT	N	NA	N
6	F	48	DM, HTN	5	Serous ca of ovary	Surgical resection plus adjuvant chemo 4m ago, 3 weeks after THBSO	N	NA	N
7	M	54	No	13	Laryngocarcinoma	4 weeks after operation	N	NA	N
8	M	80	HTN, CHD	1	Colorectal ca	3 weeks after operation and neoadjuvant chemoRT	N	NA	N
9	M	65	No	13	Esophageal ca	Surgical resection plus adjuvant chemo and RT 11m ago	Y	NA	0
10	F	70	No	124	Lung adenoca	Postoperative, recurrence. On targeted therapy	Y	NA	Y
11	F	64	No	26	Lung adenoca	Surgical resection plus adjuvant chemo, on EGFR TKI for advanced tumour	N	NA	Y
12	F	24	No	26	Breast ca	Surgical resection plus adjuvant chemo and RT 1 year ago	Y	NA	N
13	F	70	DM, HTN	148	Breast ca	Surgical resection plus adjuvant chemo and RT 12 year ago	Y	NA	N
14	F	59	No	62	Lung adenoca	Stage IV, on EGFR TKI treatment	Y	NA	Y
15	F	68	No	26	Breast ca	Surgical resection plus adjuvant chemo and RT 20m ago	N	NA	N
16	F	60	No	9	Endometrial ca	Surgical resection 7m ago, on adjuvant chemo 3 weeks ago	N	NA	N
17	M	65	No	26	Small cell lung ca	Maintenance chemo for advanced tumour	N	NA	Y
18	M	57	No	15	Lung SCC	Surgical resection plus adjuvant chemoRT 1 year ago	N	NA	N
19	F	65	No	15	Cervical ca	Surgical resection and adjuvant chemo 1 week ago	N	NA	N
20	M	47	No	51	NPC	Concurrent chemoRT 4m ago	Y	NA	N
21	M	77	No	62	Prostate ca	Postop and adjuvant RT 5 years ago, on follow up	Y	NA	N
22	M	71	DM	75	Esophageal ca	Surgical resection plus adjuvant chemo and RT 6 years ago	Y	NA	N
23	M	65	CHD	26	Prostate ca	Postoperative and adjuvant RT 2 years ago, on follow up	Y	NA	N
24	M	75	COPD	148	Rectal ca	Post-operative routine follow up	Y	NA	N
25	M	56	CHB	73	Liver ca	Surgical resection plus adjuvant chemo and RT 6 years ago, recurrence on adjuvant chemo within 4 weeks	N	NA	Y
26	M	69	No	124	Laryngocarcinoma	Postoperative and adjuvant RT 10 years ago, routine follow up	N	NA	N
27	F	52	No	8	Gastric ca	Postoperative, 4 cycles of chemo and adjuvant RT, routine follow up	N	NA	N
28	M	60	No	13	Testicular ca	Sequential chemo and RT, last cycle of chemo was within 4 weeks	Y	NA	N
Dai et al.									
No	**Gender**	**Age**	**Co-morbidities**	**Date of diagnosed cancer**	**Malignancy**	**Treatment**	**Severity (Y/N)**	**Alive (Y/N)**	**Stage IV (Y/N)**
1	M	69	Intestinal obstruction	2019.12	Peritoneal cancer	N.A	Y	Y	N
2	M	66	Jaundice, Abdominal infection, chest infection	2019.11	Gastric cancer	N.A	N	Y	Y
3	M	49	No	2020.01	Esophageal SCC	N.A	Y	Y	NA
4	M	65	No	2018.01	Gastric cancer	N.A	N	Y	N
5	M	72	Asthma	2016.09	Lung cancer	Targeted therapy (EGFR-TKI) and PD-1 immunotherapy	N	Y	Y
6	M	73	Chronic bronchitis	2019.11	Laryngeal SCC	Radiotherapy	Y	Y	Y
7	M	69	CHD, HTN	2019.08	Lung cancer	Radiotherapy	N	Y	N

8	F	55	No	2019.11	Glioblastoma	Radiotherapy and chemotherapy	N	Y	Y
9	M	74	Duodenal ulcer, COPD, HTN, CHD	2019.08	Esophageal cancer	Radiotherapy	N	Y	Y
10	M	55	Hepatic encephalopathy, GI bleeding, Chronic viral hepatitis, Anemia	2014.01	Hepatocellular carcinoma	Radiotherapy	N	Y	Y
11	M	66	HTN, DM	2019.03	Lung SCC	Chemotherapy	Y	Y	Y
12	M	67	Chronic bronchitis, HTN, Atrial fibrillation, Peptic ulcer, CKD	2019.12	Esophageal SCC	Surgery	Y	N	N
13	M	64	Cerebral infarction	2020.01	Myelodysplastic syndrome	N.A	Y	Y	NA
14	M	83	HTN	2016	Meningioma	N.A	Y	Y	NA
15	M	73	HTN	2020.01	Lung cancer	Surgery	Y	Y	N
16	F	48	No	2019.06	Breast cancer	N.A	N	Y	N
17	M	57	Prostatic hyperplasia	2016.02	Lung cancer	Radiotherapy and Targeted therapy (EGFR-TKI)	N	Y	Y
18	M	61	No	2019.12	Lung SCC	DP chemotherapy and PD-1 immunotherapy	Y	N	Y
19	F	84	HTN, DM, lumbar compression fracture	2019.06	Lung cancer	Surgery	Y	N	N
20	M	69	Abdominal aortic aneurysm	2019.09	Lung adenocarcinoma	Radiotherapy and chemotherapy	N	Y	N
21	M	69	Prosopalgia	2019.11	Lung SCC	Chemotherapy and PD-1 immunotherapy	N	Y	Y
22	M	65	Chronic viral hepatitis	2019.04	Lung cancer	N.A	N	Y	N
23	F	76	HTN, CKD, hypothyroidism, cardiac insufficiency	Unknown	Multiple myeloma	N.A	Y	N	NA
24	M	54	Incomplete ileus	2020.01	Rectal cancer	Surgery	N	Y	N
25	F	78	HTN,, COPD, CHD, CKD	2014.06	Chronic granulocytic leukemia	N.A	Y	N	N
26	M	61	HTN, DM	2018.05	Bladder cancer	N.A	Y	Y	N
27	F	50	No	2017.05	Lung cancer	N.A	N	Y	N
28	F	87	HTN	2009	Gastric cancer	N.A	Y	Y	N
29	F	55	No	2018	Cervical cancer	N.A	N	Y	N
30	F	70	HTN	Unknown	Colon cancer	N.A	N	Y	N
31	F	69	No	Unknown	Gastric cancer	N.A	N	Y	NA
32	F	59	No	2009	Breast cancer	N.A	N	Y	N
33	F	48	No	2019.06	Breast cancer	Radiotherapy	N	Y	N
34	M	57	No	2015.12	Lung adenocarcinoma	Concurrent targeted therapy（EGFR-TKI）with radiotherapy	N	Y	Y
35	M	69	Prosopalgia	2019.11	Lung SCC	Radiotherapy and Chemotherapy	N	Y	Y
36	M	69	Chronic viral hepatitis	2019.04	Lung cancer	Chemotherapy and PD-1 immunotherapy	Y	Y	Y
37	M	69	CKD	2019.09	Lung adenocarcinoma	Chemotherapy and Radiotherapy	N	Y	N
38	M	66	No	2019.08	Lung adenocarcinoma	PD-1 immunotherapy	Y	Y	Y
39	M	65	CHD	2019.04	Bladder cancer	Radiotherapy and Chemotherapy	Y	N	Y
40	M	81	HTN	2019.09	Cancer of unknown primary	PD-1 immunotherapy	Y	N	NA
41	F	46	Bacteremia	2019.01	Rectal cancer	N.A	N	Y	N
42	M	76	Intestinal obstruction	2018.11	Rectal cancer	N.A	Y	Y	Y
43	F	58	No	2020.1.2	Ovarian cancer	Chemotherapy	N	Y	N
44	F	74	DM	2020.1.12	Cervical cancer	Chemotherapy	N	Y	N
45	M	26	No	Unknown	Testicular cancer	N.A	N	Y	N
46	F	66	HTN, HLD	Unknown	Breast cancer	N.A	Y	Y	N
47	F	52	DM	Unknown	Thyroid carcinoma	N.A	N	Y	N
48	M	50	Pulmonary tuberculosis, Abdominal infection, Biliary infection, Liver abscess	2015	Thyroid carcinoma	N.A	N	Y	N
49	F	66	No	2019	Thyroid carcinoma	N.A	Y	Y	NA
50	M	83	No	2019.05	Prostate cancer	N.A	N	Y	Y
51	F	29	No	2013.05	Thyroid carcinoma	N.A	N	Y	N
52	F	32	No	2018.01	Thyroid carcinoma	N.A	N	Y	N
53	M	76	No	2018	Prostate cancer	N.A	N	Y	NA
54	F	80	HTN, HLD, Phlebothrombosis, DM	2019.01	Lung adenocarcinoma	N.A	Y	Y	N
55	M	72	HTN	2019	Lung cancer	N.A	Y	N	Y
56	F	57	No	2019.12	Breast cancer	Chemotherapy	N	Y	N
57	F	69	Chronic Hepatitis	2018	Thyroid cancer	N.A	N	Y	N
58	F	60	No	2012	Cervical cancer	N.A	N	Y	N
59	F	72	HTN,,heart disease	2013.06	Breast cancer	N.A	N	Y	NA
60	M	75	HTN	2009	Colon cancer	N.A	N	Y	NA
61	M	67	Hypothyroidism	Unknown	Liver cancer	N.A	N	Y	NA
62	F	52	HTN, CKD	2016.03	Endometrial cancer	N.A	N	Y	N
63	M	64	No	2019.08	Gallbladder carcinoma	N.A	N	Y	NA
64	M	67	No	1994	Nasopharynx cancer	N.A	N	Y	NA
65	M	63	HTN	2010	Kidney cancer	N.A	N	Y	NA
66	M	34	No	2017	Thyroid cancer	N.A	N	Y	NA
67	F	59	No	2019.07	Colon cancer	Chemotherapy	N	Y	NA
68	M	57	Cirrhosis,diabetes	2020.01	Nasopharynx cancer	Chemotherapy	N	Y	NA
69	M	62	No	2015	Bladder cancer	N.A	N	Y	NA
70	F	66	No	2015	Thyroid cancer	N.A	N	Y	NA
71	M	60	No	2015	Multiple myeloma	N.A	N	Y	NA
72	F	64	No	Unknown	Gastric stromal tumor	N.A	N	Y	NA
73	F	66	No	2004	Ocular malignancy	N.A	N	Y	NA
74	M	61	CHD, HTN	2020.01	Multiple myeloma	N.A	N	Y	NA
75	F	53	No	2019.9	Multiple myeloma	N.A	N	Y	NA
76	M	61	DM, HTN	Unknown	Malignant lymphoma	N.A	Y	N	Y
77	F	54	No	2019.03	Lung carcinoma	N.A	Y	N	N
78	F	55	No	2016	Breast cancer	N.A	N	Y	N
79	F	38	No	2019	Breast cancer	N.A	N	Y	N
80	F	50	No	2019	Rectal carcinoma	N.A	Y	N	N
81	M	67	Bronchitis	2017.12	Esophageal carcinoma	N.A	Y	Y	N
82	F	37	No	2018	Thyroid carcinoma	N.A	N	Y	N
83	F	45	No	2019.11	Cervical carcinoma	Surgery and Radiotherapy	Y	Y	N
84	M	35	No	2019.07	Glioma	N.A	Y	Y	NA
85	F	57	No	2019.11	Cervical carcinoma	Surgery	N	Y	N
86	F	49	No	2019.11	Renal carcinoma	N.A	N	Y	N
87	M	62	HTN, Tuberculosis	2018.08	Esophageal carcinoma	N.A	N	Y	N
88	M	70	CHD, Choledochectasia	2019.06	Gastric carcinoma	N.A	N	Y	N
89	M	44	No	2019.12	Thyroid carcinoma	N.A	N	Y	N
90	F	72	Cerebrovascular disease	2017.06	Breast cancer	N.A	N	Y	N
91	F	57	HTN	2019.02	Breast cancer	N.A	Y	Y	N
92	F	56	No	2019.04	Cervical carcinoma	N.A	Y	Y	N
93	M	58	No	2019.06	Rectal carcinoma	Chemotherapy	Y	Y	Y
94	M	71	CHD, HTN	2019.1	Lung SCC	Chemotherapy	Y	Y	Y
95	F	67	DM HTN	2019.12	Breast cancer	Surgery	Y	Y	N
96	M	70	No	2019.11	gastric carcinoma	Chemotherapy	Y	Y	N
97	M	70	HTN	2014	Esophageal carcinoma	N.A	Y	Y	N
98	F	42	HTN	2020.01	Pancreatic cancer	Surgery	N	Y	N

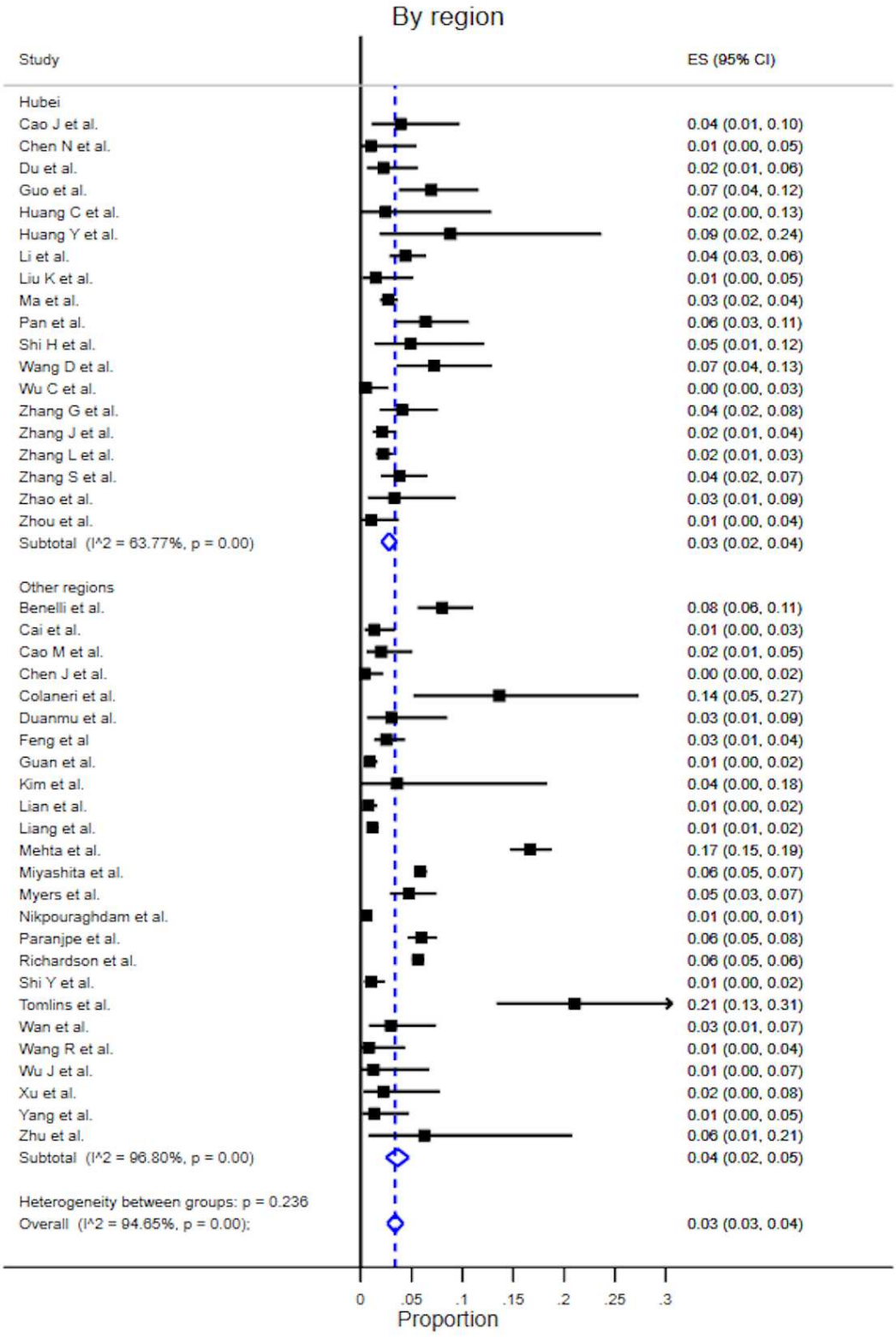
99	M	43	No	2011	gastric carcinoma	N.A	Y	Y	N
100	M	69	No	2015	Lung cancer	N.A	Y	Y	N
101	M	62	Cerebral hemorrhage, Uremia, DM, HTN	2020.01	Lung cancer	N.A	N	Y	N
102	M	37	HLD	2018.06	Sigmoid colon cancer	N.A	Y	Y	N
103	F	70	CHD, HTN	2002	Thyroid cancer	N.A	Y	Y	N
104	M	51	No	2015	Leukemia	Targeted therapy (Imatinib Mesylate)	N	Y	N
105	F	55	HTN	2019.07	Nasopharyngeal lymphoma	N.A	N	Y	N

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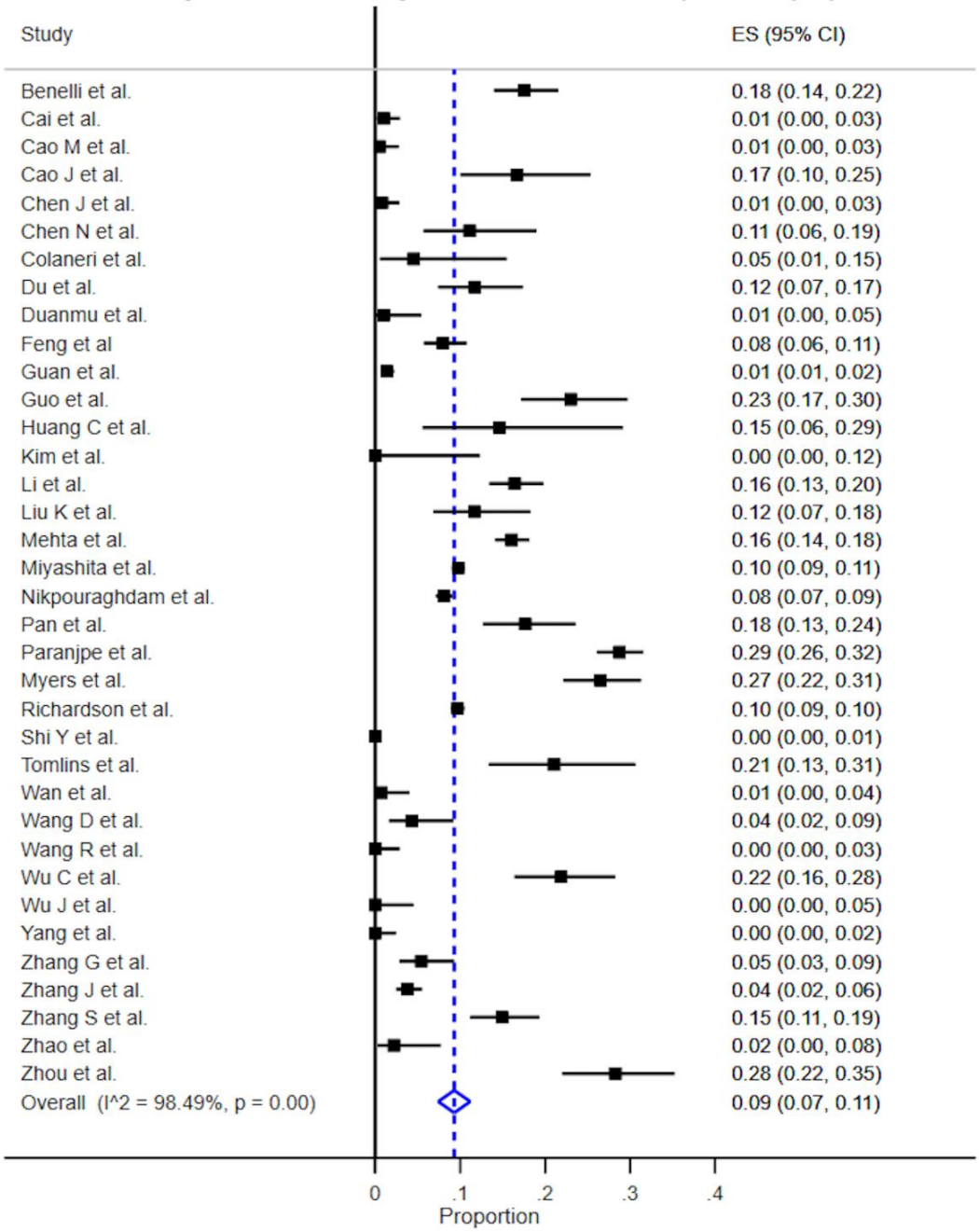
No	Gender	Age	Co-morbidities	Length of dx of cancer in months	Malignancy	Treatment	Severity (Y/N)	Alive (Y/N)	Stage IV (Y/N)
1	F	48	No	46	Breast	RT+endocrinotherapy	N	Y	NA
2	M	57	No	53	Lung	targeted therapy/RT	N	Y	NA
3	M	65	Liver cirrhosis	50	Lung	chemotherapy+immunotherapy	N	Y	NA
4	M	66	No	53	Lung	chemotherapy+immunotherapy+targeted therapy	N	Y	NA
5	M	69	No	48	Lung	chemotherapy+RT+immunotherapy	N	Y	NA
6	M	69	No	32	Lung	chemotherapy+RT	Y	N	NA
7	M	61	No	36	Urinary	chemotherapy	Y	N	NA
8	M	37	No	36	liver	surgery+chemotherapy	N	Y	NA
9	M	87	COPD, HTN, CHD, cerebrovascular disease/chronic nephrosis/immunological diseases	44	Gastric	follow-up	Y	N	NA
10	M	61	Disturbance of consciousness	16	Lung	chemotherapy+immunotherapy	Y	N	NA
11	F	67	DM, HTN, liver cirrhosis	8	Esophagus	perioperative	Y	N	NA
12	M	65	HTN, cerebrovascular disease	48	Lung	follow-up	N	Y	NA
13	F	50	No	54	HN	follow-up	N	Y	NA
14	F	46	HTN	50	Thyroid	follow-up	N	Y	NA
15	M	77	HTN	42	Colorectal	follow-up	Y	Y	NA
16	F	70	No	34	Breast	follow-up	Y	Y	NA
17	F	75	No	51	Colorectal	follow-up	N	Y	NA
18	F	53	No	57	HN	follow-up	N	Y	NA
19	M	48	HTN	49	CNS	follow-up	N	Y	NA
20	F	58	HTN, HLD, CHD	45	Thyroid	follow-up	N	Y	NA
21	M	54	No	44	Colorectal	perioperative	N	Y	NA
22	F	78	HLD	18	Hematology	supportive treatment	Y	N	NA
23	M	60	HTN	64	Hematology	supportive treatment	N	Y	NA
24	F	83	HTN	64	CNS	follow-up	Y	Y	NA
25	M	68	HTN	47	Urinary	follow-up	N	Y	NA
26	M	66	COPD, HTN, CHD, liver cirrhosis	48	Urinary	follow-up	N	Y	NA
27	M	63	HTN	29	Urinary	follow-up	Y	N	NA
28	M	77	No	55	Urinary	radiotherapy	Y	Y	NA
29	M	73	HTN/HLD/immune disease	53	Lung	perioperative	N	Y	NA
30	M	66	HD/DM/CHD/immune disease	22	Duodenal	supporting treatment	Y	Y	NA
31	F	84	HTN, immune disease	11	Lung	perioperative	Y	N	NA
32	M	67	DM, HTN, CHD, chronic nephrosis	42	Colorectal	follow-up	Y	Y	NA
33	F	65	DM, HTN, liver cirrhosis	41	CNS	follow-up	N	Y	NA
34	M	78	HTN	40	Colorectal	follow-up	N	Y	NA
35	M	65	No	38	Colorectal	perioperative	Y	Y	NA
36	F	71	Cerebrovascular disease	37	Colorectal	follow-up	Y	Y	NA
37	F	65	HTN, tuberculosis	44	Breast	endocrinotherapy	Y	Y	NA
38	F	40	No	41	Thyroid	follow-up	N	Y	NA
39	F	61	HTN	47	Thyroid	follow-up	N	Y	NA
40	F	70	COPD, DM, HTN, cerebrovascular disease	6	Breast	follow-up	Y	N	NA
41	M	98	No	21	Colorectal	follow-up	Y	N	NA
42	M	64	HTN	53	Esophagus	chemotherapy	N	Y	NA
43	M	67	No	14	Urinary	follow-up	Y	N	NA
44	F	76	COPD, CHD, HTN	52	Hematology	follow-up	N	Y	NA
45	F	68	Tuberculosis	53	Thyroid	follow-up	N	Y	NA
46	M	66	HTN	50	Lung	follow-up	N	Y	NA
47	M	69	No	47	Urinary	follow-up	Y	Y	NA
48	M	69	Tuberculosis	44	Colorectal	follow-up	N	Y	NA
49	M	70	DM, HTN	48	Gastric	follow-up	Y	Y	NA
50	M	72	No	19	Lung	chemotherapy	Y	N	NA
51	F	55	HTN	48	Lung	follow-up	N	Y	NA
52	F	58	No	50	CNS	follow-up	N	Y	NA
53	M	54	HTN	50	HN	follow-up	N	Y	NA
54	F	64	No	40	Breast	follow-up	N	Y	NA
55	F	51	DM, CHD	52	Thyroid	follow-up	N	Y	NA
56	M	56	HTN	50	Thyroid	follow-up	Y	Y	NA
57	M	71	No	13	HN	follow-up	Y	N	NA
58	F	65	DM, HTN	36	Ovary	follow-up	N	Y	NA
59	M	90	HTN	9	Colorectal	perioperative	Y	N	NA
60	F	64	DM, HTN	54	Breast	follow-up	N	Y	NA
61	F	69	No	54	Lung	follow-up	Y	Y	NA
62	M	52	No	39	Thyroid	follow-up	Y	N	NA
63	M	74	HTN	22	Colorectal	follow-up	Y	N	NA
64	M	63	HTN	23	HN	follow-up	Y	N	NA
65	M	70	DM, HTN	37	Lung	follow-up	N	Y	NA
66	M	67	HTN	21	Lung	immunotherapy	Y	N	NA
67	M	64	HTN	57	Urinary	follow-up	Y	Y	NA

Abbreviations: BSC: best supportive care; ca – carcinoma, CHB: chronic hepatitis b; CHD: coronary heart disease; CKD: chronic kidney disease; COPD: chronic obstructive pulmonary disease; DM: diabetes mellitus; EGFR: epidermal growth factor receptor; HLD: hyperlipidemia; HTN: hypertension; NPC: nasopharyngeal cancer; NSCLC: non-small cell lung cancer; PD1: programmed death 1; RCC: renal cell cancer; RT: radiotherapy; TKI: tyrosine kinase inhibitor; SCC: squamous cell cancer; THBSO: total hysterectomy and bilateral salpingo-oophrectomy
Y: yes; N: no; NA: not applicable/available

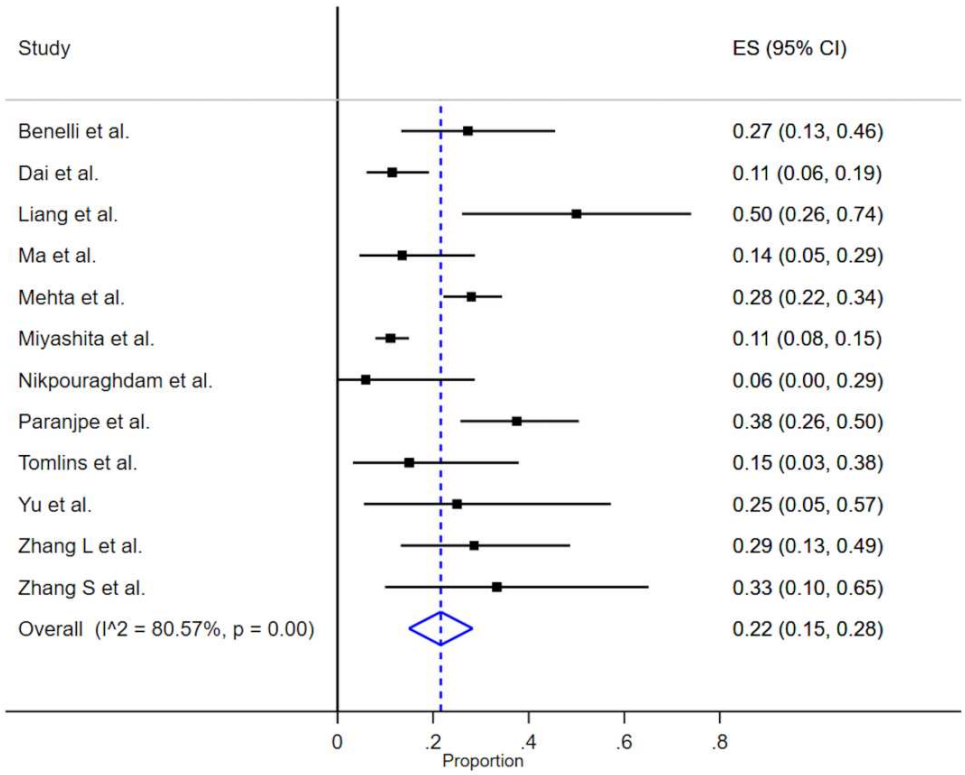
Supplementary Figure 1: Random effects pooled proportion of cancer patients among total patients diagnosed with COVID-19 (subgroup by region – within Hubei province and outside Hubei province)



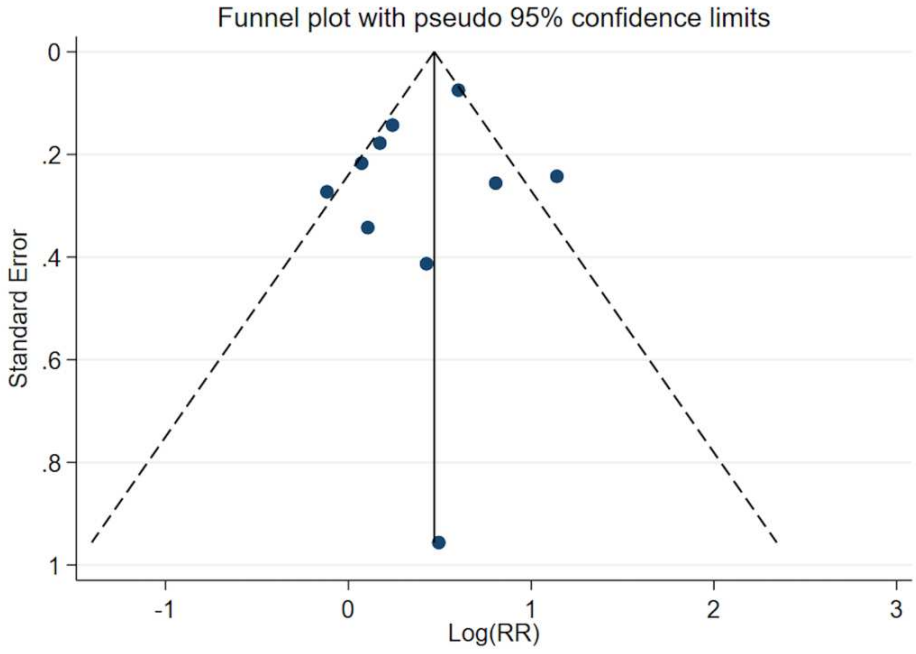
Supplementary Figure 2a: Random effects pooled proportion of deaths among patients diagnosed with COVID-19



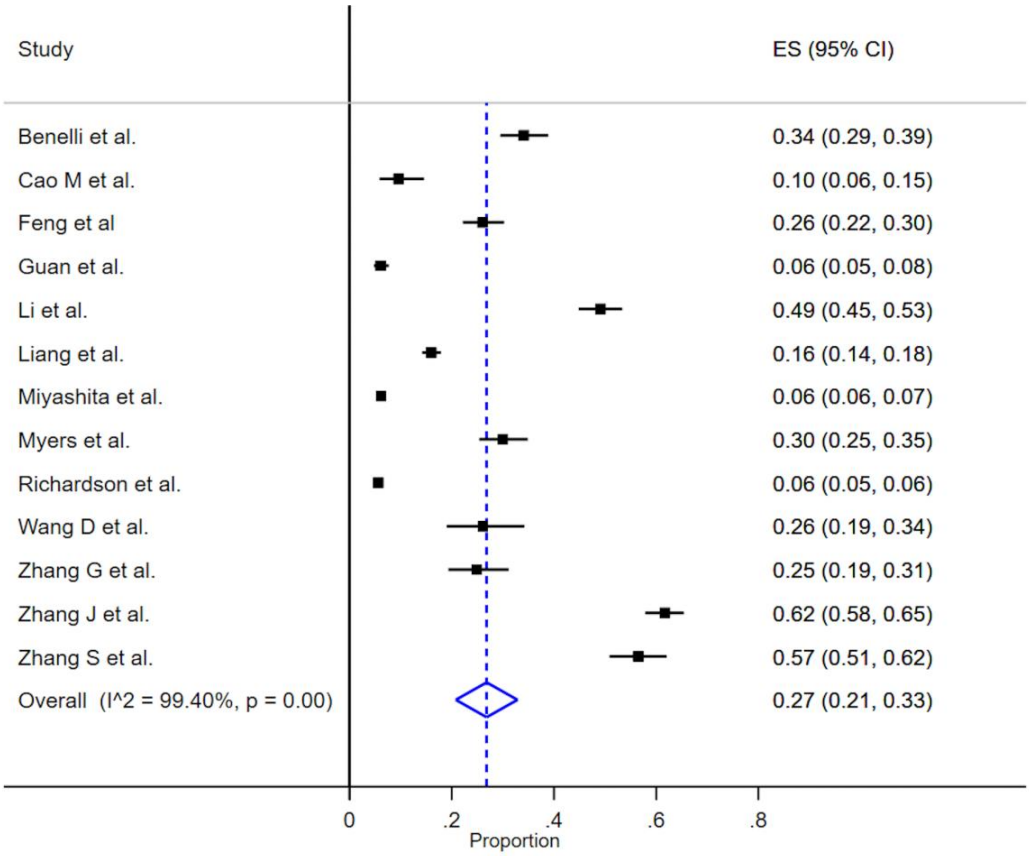
Supplementary Figure 2b: Random effects pooled proportion of deaths among cancer patients diagnosed with COVID-19



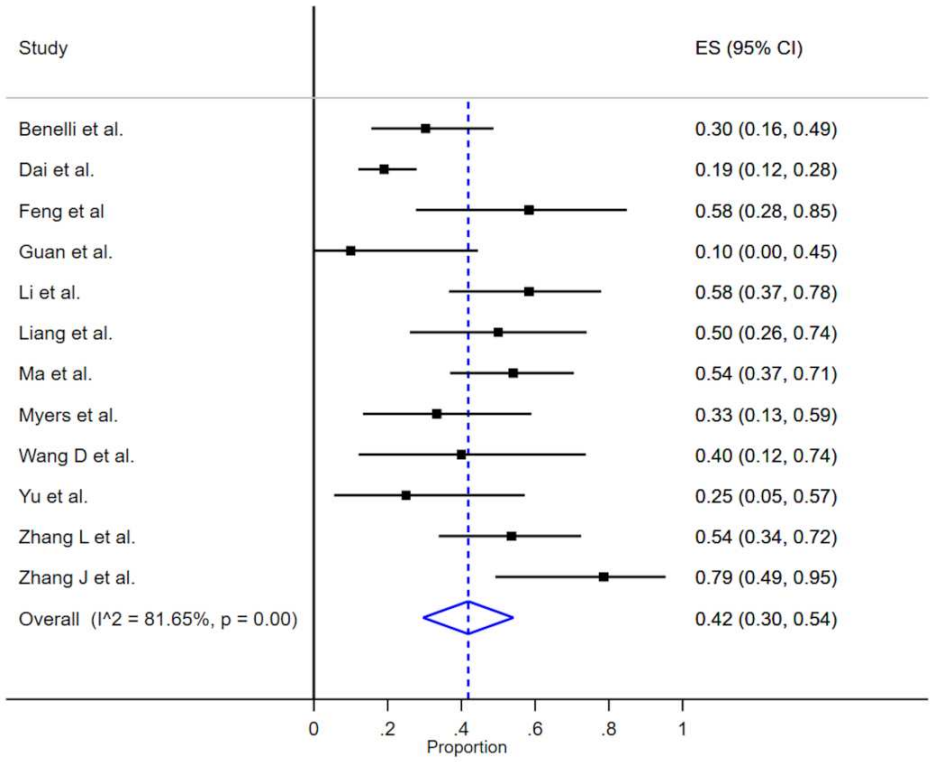
Supplementary Figure 2c: Funnel plot of ten case series included in pooled relative risk analysis of death outcomes between cancer and general COVID-19 patients



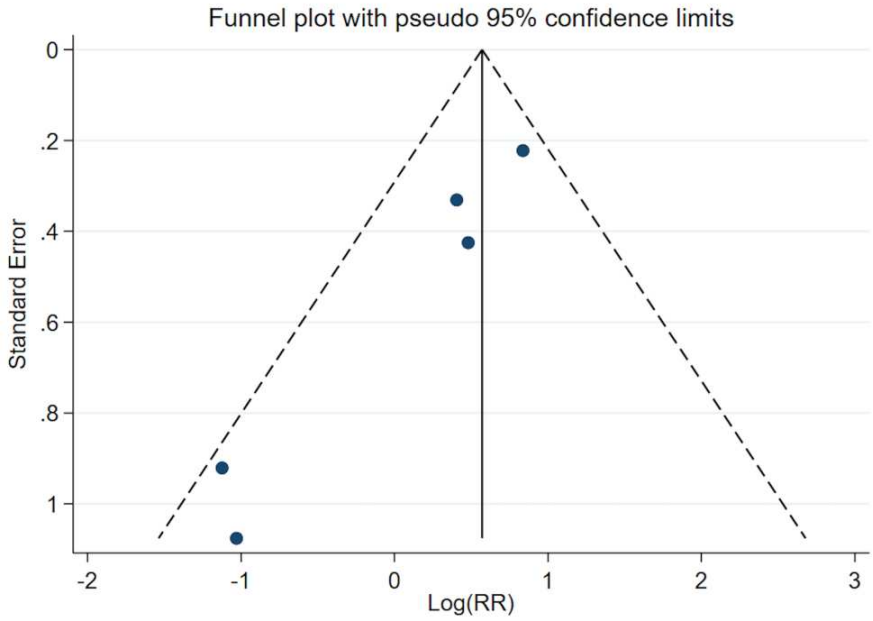
Supplementary Figure 3a: Random effects pooled proportion of severe symptoms among patients diagnosed with COVID-19



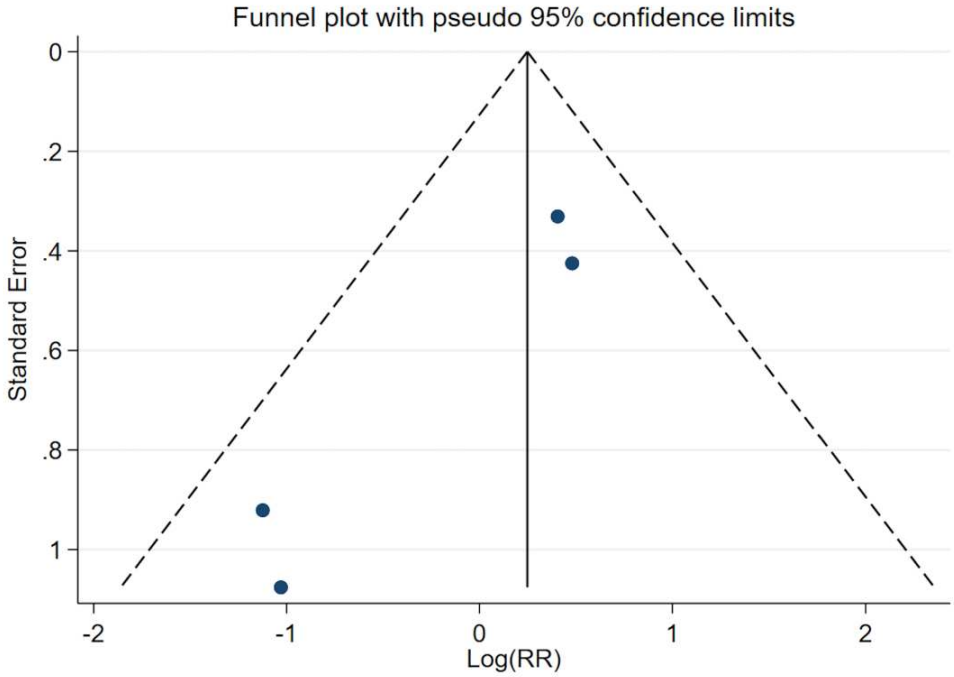
Supplementary Figure 3b: Random effects pooled proportion of severe symptoms among cancer patients diagnosed with COVID-19



Supplementary Figure 3c: Funnel plot of five case series included in pooled relative risk analysis of severe outcomes between cancer and general COVID-19 patients



Supplementary Figure 4: Funnel plot of four case series included in pooled relative risk analysis of lung cancer vs non-lung cancer and stage IV vs non stage IV cancers



Supplementary Table 2: Characteristics of 230 cancer patients with COVID-19 infection with severe and non-severe outcomes

COVID-19 severity	Non-severe (N = 131)	Severe (N = 99)	P-value (Chi-square)
Gender			
Male	70 (50.0)	70 (50.0)	0.010
Female	61 (67.8)	29 (32.2)	
Co-morbidities			
≥2	20 (8.7)	25 (10.9)	0.093
0-1	102 (44.3)	71 (30.9)	
Stage			
Stage IV	20 (8.7)	20 (8.7)	0.635
Stage I-III	56 (24.3)	40 (17.4)	
Unknown	55 (23.9)	39 (17.0)	
Tumour Type			
Lung cancers	32 (13.9)	24 (10.4)	0.974
Non-lung cancers	99 (43.0)	75 (32.6)	
Age			
≥ 60	63 (27.4)	79 (34.3)	<0.001
< 60	59 (25.7)	17 (7.4)	
Treatment			
Active systemic therapy	33 (14.3)	23 (10.0)	0.758
No systemic therapy	97 (42.2)	76 (33.0)	